

Mall Customer Segmentation

Problem Statement & Objective

The purpose of this project is to analyze customer behavior and purchasing patterns in a supermarket mall using membership card data. The dataset contains customer ID, age, gender, annual income, and spending score (a value assigned based on purchasing behavior).

The mall's goal is to understand its customer segments to better target marketing efforts and increase sales. The objective is to use **unsupervised machine learning techniques** to identify distinct customer groups and provide actionable recommendations for marketing strategies.

Part 1: Modeling, Hyperparameter Tuning & Evaluation

Data Preparation & EDA

- Imported libraries: pandas, numpy, matplotlib, seaborn, pycaret, scikit-learn, LIME, SHAP.
- Dataset: mall_customers.csv
- Initial exploration: checked shape, head/tail, info, describe.

Exploratory Data Analysis

- **Age Distribution:** Majority of customers between 30–40 years. Senior customers are fewer.
- **Annual Income Distribution:** Most customers earn between 50k–70k.
- **Spending Score:** Concentrated around 40–60, indicating moderate spending.
- **Gender Distribution:** More females than males.

Key insight: **Younger customers spend more, older customers spend less.**

Model Training with PyCaret

Tested clustering models:

1. KMeans

- Determined optimal number of clusters with elbow method (6 clusters).
- Visualized in 3D (Age, Income, Spending Score).
- Clusters were reasonably well-separated.

2. Hierarchical Clustering (HClust)

- Optimal clusters = 6
- Produced slightly better separation than KMeans.

3. DBSCAN

- Some data points overlapped, weaker cluster definition.

Evaluation Metrics (No. of clusters = 6)

Model	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
KMeans	0.42	141	0.86
HClust	0.44	159	0.76
DBSCAN	0.1845	34.0705	1.7569

Interpretation of Metrics:

- Silhouette Score: Higher is better. HClust (0.44) > KMeans (0.42).
- Calinski-Harabasz: Higher is better. HClust (159) > KMeans (141).
- Davies-Bouldin: Lower is better. HClust (0.76) < KMeans (0.86).

Best Model: Hierarchical Clustering (HClust) performed best across metrics and visualization.

Hyperparameter Tuning

- Tuned number of clusters.
- Optimal: **6 clusters**.

Part 2: Model Explanation & Understanding

Approach 1: Manual Cluster Analysis

For each cluster, computed mean values of Age, Income, and Spending Score. Assigned meaningful names:

Cluster	Avg Age	Avg Income (k\$)	Avg Spending Score	Segment Name
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0	27.38	57.51	45.84	Young Moderate Income & Moderate Spenders
1	56.40	55.29	48.36	Older Moderate Income & Moderate Spenders
2	32.69	86.54	82.13	Young High Income & High Spenders
3	43.89	91.29	16.68	Middle-Aged High Income & Low Spenders
4	44.32	25.77	20.27	Middle-Aged Low Income & Low Spenders
5	24.81	25.62	80.24	Young Low Income & High Spenders

Approach 2: Classification + Explainability (SHAP & LIME)

- Trained a **Random Forest Classifier** using cluster labels as the target.
- Achieved **accuracy = 1.0**, due to clusters being derived from the same features (Age, Income, Spending).

SHAP (Global Explanation)

- **Spending Score**: Most influential feature across clusters.
- **Annual Income**: Second most important, strongly separates clusters.
- **Age**: Lowest impact, but differentiates older vs. younger groups.

LIME (Local Explanation)

Example: Customer with Age < 25, Income > 60k, Spending > 70 → classified as **“Young High Income & High Spenders.”**

- Spending Score: +0.45 (pushed to cluster)
- Income: +0.30 (pushed to cluster)
- Age: -0.10 (weaker impact)

Comparison of Approaches

- **Manual analysis** gave intuitive cluster names that marketing teams can directly use.
- **Model explainability (SHAP & LIME)** validated that the clusters are meaningful and based on logical feature contributions.
- Both approaches align: clusters are interpretable and trustworthy.

Final Model & Deployment

- Final model: **Hierarchical Clustering (6 clusters)**.
- Retrained on the full dataset (train + test).
- Exported as `Datadivr\mall_customer_segmentation.pkl`.

Observations & Outcomes

1. **Best Algorithm:** Hierarchical Clustering outperformed KMeans and DBSCAN in separation and metrics.
2. **Customer Segments Identified:** Six well-defined clusters with meaningful behavioral names.
3. **Explainability:** SHAP and LIME confirmed that Spending Score and Income drive segmentation, while Age provides secondary separation.
4. **Business Value:** The mall can now:
 - Target **Young High Income & High Spenders** with premium product marketing.
 - Offer discounts to **Older High Income & Low Spenders** to increase purchases.
 - Design affordable bundles for **Young Low Income & Low Spenders**.

Why These Approaches Were Taken

- **PyCaret** accelerated experimentation with multiple clustering algorithms.
- **HClust** chosen for superior performance and clear separation.
- **Manual + Explainable AI** approaches ensured clusters are **interpretable, business-relevant, and trustworthy**.
- Modular code design ensures reusability and efficient experimentation.

Conclusion

The project successfully segmented mall customers into six actionable groups using unsupervised learning. The final model, combined with explainability techniques, provides both **data-driven segmentation** and **human-friendly interpretation**. This enables the marketing team to develop **targeted campaigns** that can directly improve mall revenue.